

PRASHANTH BHARGAV NUTALAPATHI

Hyderabad, India | +91-8332829717 | +91-9398952415 | pbnutalapathi@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML Engineer with around 3 years building and deploying production AI/ML systems. Developed GPU-accelerated computer vision pipelines for drone-based inspection using PyTorch, YOLO, Faster R-CNN, and SAM, deployed via Flask APIs and Docker. Hands-on GenAI engineering across RAG pipelines, agentic LLM workflows, and edge AI deployment using LangChain, LangGraph, FAISS, and Azure OpenAI.

WORK EXPERIENCE

DigiTele Networks – AI/ML Engineer (Junior Data Analyst), Hyderabad Dec 2022 – July 2025

- **Computer Vision Pipelines:** Built CUDA-accelerated wind turbine inspection pipelines (YOLO, Faster R-CNN, ~80%+ mAP) and multi-spectral (Thermal & RGB) drone surveillance models (~75-80% mAP) using PyTorch and OpenCV.
- **Training Data Pipelines:** Designed training data pipelines using Roboflow for annotation, augmentation, and confidence-based hard negative filtering as an active learning strategy.
- **Distributed Inference:** Leveraged Ray Core for parallelized preprocessing and distributed inference across multiple computer vision models, enabling scalable defect detection on large drone image datasets.
- **Post-processing:** Built pipelines integrating SAM-based segmentation with camera metadata (GPS, altitude, focal length) to map pixel-level detections to real-world measurements; applied custom NMS and confidence thresholding for improved prediction accuracy
- **ML Inference Services:** Built REST APIs using Flask for real-time and batch inference; containerized with Docker for scalable deployment.
- **Edge AI & Hardware Optimization:** Exported YOLOv5 to ONNX (validated via ONNX Runtime) and TFLite with INT8 post-training quantization (~75% size reduction), benchmarking latency on Qualcomm QRB5.

Techniq Design Group – Trainee Engineer, Hyderabad Nov 2018 – June 2019

- **Embedded Driver Development:** Developed and tested Embedded C drivers for UART, I2C, and SPI communication protocols on microcontroller-based systems.
- **Hardware Design:** Designed PCB layouts using KiCad and performed precision component placement and soldering for electronic boards.

Bharat Forge Ltd – Diploma Engineer Trainee, Hyderabad Feb 2017 – Feb 2018

- **Systems Integration:** Assembled defense-grade hardware and ECU systems, ensuring compliance with IPC/WHMA-A-620 Class 3 standards.
- **Validation & Testing:** Conducted functional and environmental testing using custom test jigs to identify hardware issues and support design validation

EDUCATION

MLR Institute of Technology - B.Tech in Electronics and Communication Engineering (Lateral Entry) - 2019–2022

Central Institute of Tool Design - Diploma in Automation and Robotics Engineering - 2013–2016

PROJECTS

- **DocSense Chatbot (Azure OpenAI · Azure Blob Storage · Azure Container Apps · LangChain · ChromaDB · BM25 · Docling)**
 - ▶ Built a RAG chatbot with LangChain, Azure OpenAI (text-embedding-3-small, GPT-4.1-mini) and ChromaDB, combining vector search with BM25 for hybrid retrieval over PDFs ingested with Docling HybridChunker.
 - ▶ Exposed the pipeline via FastAPI for document upload and querying; stored raw documents in Azure Blob Storage, vectors in ChromaDB on Azure Files, deployed on Azure Container Apps.
- **Loan Assessment AI Chatbot (XGBoost · SHAP · ChromaDB · BM25 · MMR · Gemini API)**
 - ▶ Performed EDA, statistical feature analysis, and XGBoost modeling with scale_pos_weight; tracked experiments using MLflow and explained model predictions with SHAP.
 - ▶ Developed a hybrid RAG pipeline with recursive chunking, BAAI/BGE embeddings, and BM25 + MMR retrieval over ChromaDB for policy-grounded responses.
 - ▶ Integrated Gemini API for structured loan approval explanations; leveraged LangSmith tracing to monitor EnsembleRetriever performance.

SKILLS

Languages & ML: Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP, NLP, OpenCV

GenAI & RAG: LangChain, LangGraph, RAG, Agentic Workflows, Prompt Engineering, ChromaDB, FAISS, HuggingFace, Groq, Ollama

Tools & Cloud: Azure (OpenAI, Blob Storage, Container Apps, Files), FastAPI, Flask, Docker, Ray, Git, MongoDB, Pandas, NumPy, MLflow, LangSmith